Omar Ayman Bakr

Machine Learning Engineer

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Portfolio LinkedIn Github LeetCode

Mechatronics Engineering graduate specializing in Machine Learning and Computer Vision, with 500+ LeetCode problems solved. Currently building a retrieval-augmented document Q&A service end to end.

TECHNICAL SKILLS

- **Languages:** Python, C++, C, MATLAB, SQL
- **ML & Deep Learning:** PyTorch, Scikit-learn, LangChain, XGBoost, LightGBM, NumPy, Pandas, Matplotlib
- **Backend & Data:** FastAPI, MongoDB, PostgreSQL, Docker, Git, Linux/Fedora, REST APIs, async I/O
- **Foundations:** Data Structures & Algorithms, Linear Algebra, Probability, Calculus

PROJECTS

NotebookLLM-minus — Retrieval-Augmented Document Q&A Service *(in progress)*

- Building a NotebookLM-style backend on FastAPI and MongoDB: documents are uploaded per project, chunked with LangChain, and stored for grounded question answering; embedding and vector retrieval in active development.
- Designed a layered service (routes → controllers → models) around a single error boundary — typed exceptions carry their own HTTP status — with async Motor I/O and correlation-id logging on every request.

Volleyball Group Activity Recognition

- Built eight progressively complex baselines on the Volleyball Dataset (55 videos, 4,830 clips, 60+GB) in PyTorch, isolating one architectural component at a time from a single-frame ResNet-50 to hierarchical player- and scene-level LSTMs.
- Best model reaches **85.6% accuracy, 0.86 macro-F1**, exceeding the CVPR 2016 published result of 81.9%.

Credit Card Fraud Detection on Severely Imbalanced Data

- Benchmarked 11 models on a 285K-transaction dataset with a $< 0.2\%$ fraud rate, implementing Focal Loss from its mathematical definition in PyTorch to attack the imbalance algorithmically rather than by resampling.
- XGBoost reached **F1 0.89, ROC-AUC 0.98** on the minority class — a 16% F1 gain over the logistic-regression baseline.

New York City Taxi Trip Duration Prediction

- Engineered 55 features from 10 originals by integrating five external sources (NYC DOT traffic, weather, borough geometry, holidays) and deploying a local OSRM routing server in Docker for real-world distance and travel time on 1M+ trips.
- Ridge Regression achieved **RMSE 0.397, R^2 71.6%** on 224,734 held-out trips.

EDUCATION

Bachelor of Science, Mechatronics Engineering 2019 – 2024
Helwan University — Graduated with Honors (Very Good, GPA 3.0/4.0)

- Built a mathematical model in MATLAB for a green hydrogen generator, controlling input voltage and current while predictively monitoring safety parameters such as operating temperature.